

Lab 1

Requirements Engineering & UML Use-Case Modelling

Household Carbon Footprint & Target Tracker

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1. Problem Context

Household Carbon Footprint & Target Tracker The green living analytics is a tool through which households record monthly electricity and gas use and kilometres travelled for commuting to and from work each month to provide CO equivalents for tracking carbon reduction milestones.

Target Actors

Resident - given in the problem statement

Sustainability Coach – state it in the Problem Statement.

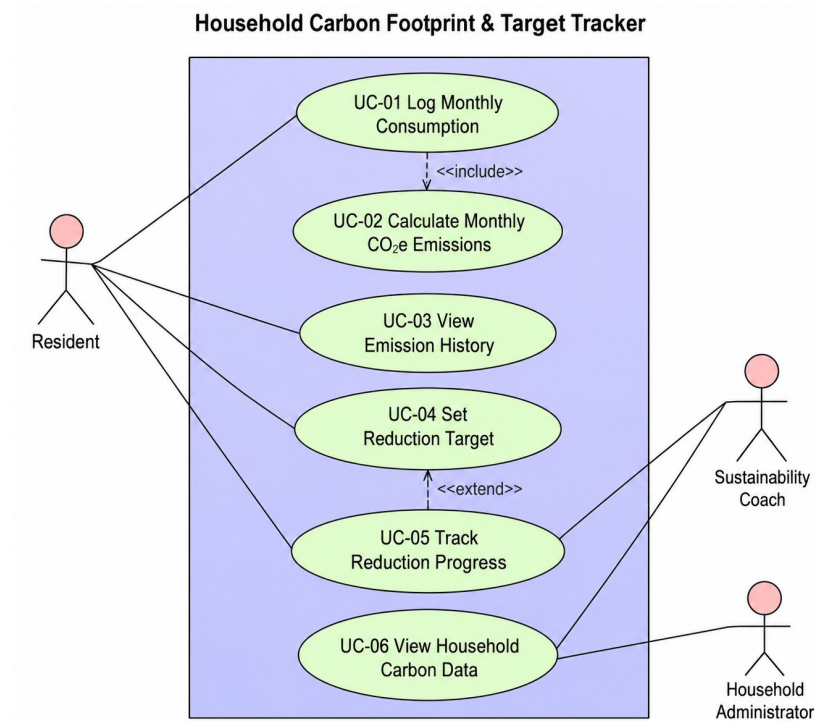
Household Administrator - included to meet the condition that there be a minimum of three actors

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow Residents to enter monthly electricity consumption, gas consumption, and commute distance.	High	Pass: Valid electricity, gas, and commute values are accepted and stored. Fail: Any valid consumption value cannot be entered or stored.	Provides the input data required to calculate the household carbon footprint.
FR-002	Functional	The system shall calculate the household's total monthly CO ₂ e emissions from the recorded electricity, gas, and commute data.	High	Pass: The system calculates and displays the monthly CO ₂ e value after valid data is submitted. Fail: The system does not calculate or display the value.	Core functionality for measuring household carbon emissions.

FR-003	Functional	The system shall display historical monthly CO ₂ e emissions for the household.	Medium	Pass: Previously recorded monthly CO ₂ e values are displayed in chronological order. Fail: A previously recorded month's emissions cannot be displayed.	Allows Residents to track changes in emissions over time.
FR-004	Functional	The system shall allow Residents to set a carbon-reduction target and track progress towards the target.	High	Pass: A reduction target can be saved and progress towards it is displayed. Fail: The target cannot be saved or progress is not displayed.	Supports tracking of household carbon-reduction milestones.
FR-005	Functional	The system shall allow the Sustainability Coach to view household carbon-footprint data and reduction progress.	Medium	Pass: The Sustainability Coach can view the household's recorded emissions and target progress. Fail: The required household information cannot be accessed.	Enables monitoring and guidance of household sustainability progress.
NFR-001	Non-functional – Performance	The system shall render annual carbon-reduction milestone charts and historical comparison graphs in under 200 ms.	High	Pass: The chart renders in less than 200 ms during benchmark testing. Fail: Rendering takes 200 ms or longer.	Ensures responsive visualization of carbon-reduction progress.
NFR-002	Non-functional – Security	The system shall restrict access to household carbon-footprint data to authorized users only.	High	Pass: Unauthorized users are denied access to household carbon-footprint data. Fail: An unauthorized user can access the data.	Protects household consumption and emissions information.

3. UML Use-Case Diagram

System: Household Carbon Footprint & Target Tracker



Actors:

- Resident
- Sustainability Coach
- Household Administrator

Use Cases:

- **UC-01:** Log Monthly Consumption
- **UC-02:** Calculate Monthly CO₂e Emissions
- **UC-03:** View Emission History
- **UC-04:** Set Reduction Target
- **UC-05:** Track Reduction Progress
- **UC-06:** View Household Carbon Data

«include»:

UC-01 Log Monthly Consumption → UC-02 Calculate Monthly CO₂e Emissions

«extend»:

UC-05 Track Reduction Progress → UC-04 Set Reduction Target

4. Use-Case Flow Specification

Use Case: UC-01 – Log Monthly Consumption

Primary Actor: Resident

Preconditions

1. Resident is logged into the system.
2. The household profile exists.
3. Resident has permission to enter household consumption data.

Postconditions

4. Monthly consumption data is recorded successfully.
5. Monthly CO₂e emissions are calculated.
6. The updated carbon footprint is available for tracking.

Main Success Scenario

7. Resident selects the household's monthly consumption entry option.
8. System displays the monthly consumption form.
9. Resident enters electricity consumption in kWh.
10. Resident enters gas consumption.
11. Resident enters commute distance in kilometres.
12. System validates the entered values.
13. System calculates the household's monthly CO₂e emissions.
14. System stores the consumption data and calculated emissions.
15. System displays the updated monthly CO₂e value.
16. Use case ends successfully.

Alternate Flow

7a. Invalid Consumption Data

17. System detects an invalid or incomplete consumption value.
18. System displays an error message.
19. Resident is prompted to correct the entered value.
20. Resident submits the corrected data.
21. The flow resumes from Step 6 of the main scenario.